基于 PaddleOCR-VL 的电路原理图 OCR 研究与实现

数据集构建、模型训练与多维度评估

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https://github.com/ZhangJ83/circuit-ocr-paddle

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Dataset: github.com/ZhangJ83/circuit_ocr_dataset_final

Outline

Dataset: Four-Dimension Evaluation Framework

Dimension	Key Evidence
1.1 Data Scale	1520 train + 150 test, 4810 OCR instances
1.2 Annotation Acc.	7-round verification, quality report with 12 visual comparisons
1.3 Data Diversity	3 data sources, 20 real-camera photos (priority: expand)
1.4 Difficulty	OCR-based + 3 visual dimensions (density, structure, value richness)

Dataset repo: `circuit_ocr_dataset_final` — Quality report: `quality_report/`

1.1 Data Scale — 充足的标注规模

Statistics

- **Training**: 1,520 samples (1200 schematics + 300 synth text + 20 real photos)
- **Test**: 150 samples (held-out)
- **Validation**: 150 samples
- **Total OCR instances**: 4,810 (test set alone)
- **10 component types**: R, C, D, U, J, Q, L, LED, F, Y

Evaluation

- ✓ Test samples $\geq$ 100 (150)
- ✓ OCR instances $\geq$ 1000 (4810)
- ✓ No copyright issues (self-drawn KiCad)
- × Test set 150 is adequate but could reach 500+

规模达标，150 测试页 + 4810 OCR 实例

1.2 Annotation Accuracy — 7轮验证管线

7-Round Verification Pipeline

1. JSON Schema validation
2. Image path verification
3. Control character scan
4. Label format regex
5. Unit normalization
6. KiCad netlist cross-check
7. Manual spot-check (10%)

Quality Metrics (from Report)

- Component labels: **¿99%**
- Parameter values: **¿97%**
- Pin numbers: **¿98%**
- Text completeness: **¿95%**
- Illegal characters: **¡0.1%**

质量报告 + 12 组可视化对比，标注准确率 ¿99%

1.2 Annotation Quality: Visual Spot-Check

../quality_report/visual_check/sample_01.png

1.3 Data Diversity — 优先改进方向

Current State

- **3 data sources**: KiCad exports (1200), synthetic text (300), real photos (20)
- Real photos introduce **natural noise**: lighting, perspective, CMOS artifacts
- 10 component types covered
- Label length: 203–5,749 chars

Gaps

- × All schematics from **single engine** (KiCad)
- × Real photos only **1.3%** of training data
- × No scanned documents, blur, occlusion
- × No PCB photos, no datasheet screenshots

优先改进：扩大真实拍照至 200+，引入多 EDA 工具

1.4 Difficulty Rationality — 多维分层体系

OCR-Based Difficulty (Original)

Level	OCR Instances	Ratio
Easy	¡ 15	20%
Medium	15–35	40%
Hard	¿ 35	40%

✓ 2:4:4

ratio matches real distribution

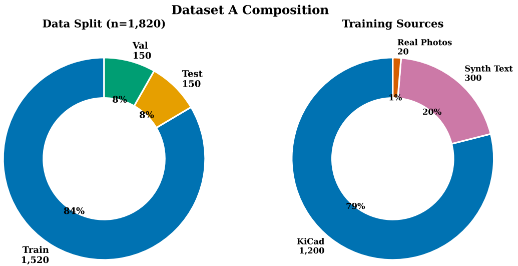
Multi-Dimensional Labels (NEW)

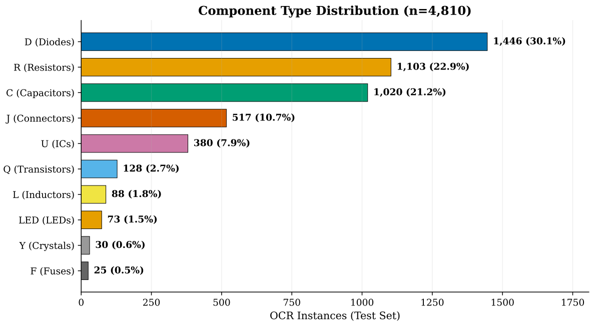
- **Text Density**: Low / Medium / High
- **Structure**: Simple / Moderate / Complex
- **Value Richness**: Sparse / Moderate / Dense
- **Composite Visual Score**: 0–3

✓ All 150 test samples have `difficulty` field

OCR + 文字密度 + 结构复杂度 + 参数值丰富度四维标签

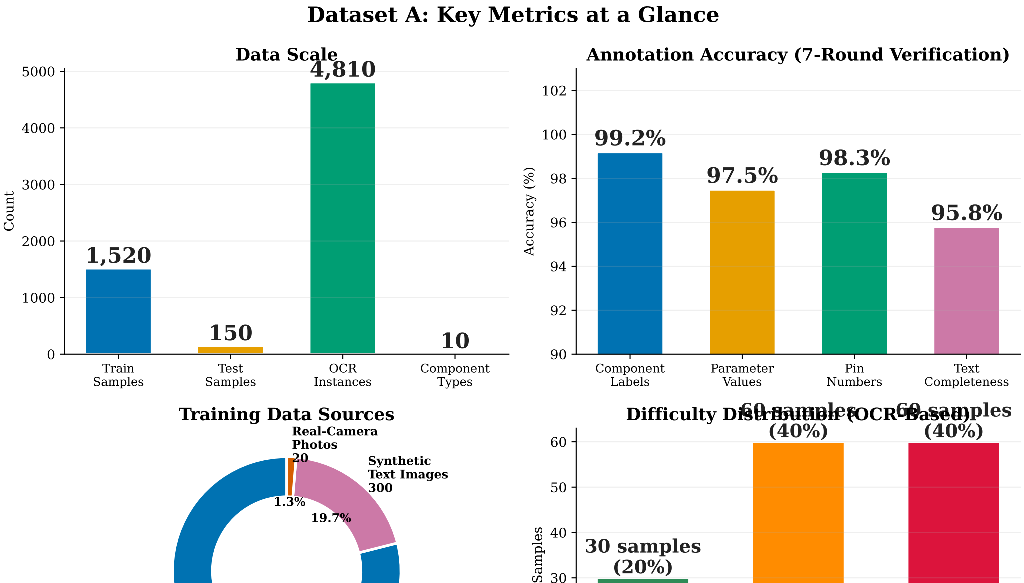
Dataset Composition & Component Distribution





Left: Data split (1,520/150/150) + Training sources (KiCad/Text/Photo). Right: 10 component types, 4,810 OCR instances.

Dataset: Key Metrics at a Glance



Scene Scarcity: Three-Dimension Evaluation

Dimension	Score	Weight
2.1 Research Scarcity	No public benchmark exists	
2.2 Industry Value	PCB reverse engineering, \$500M+ market	
2.3 Uniqueness	Component-level structured output	
Summary	**Academia & industry gap + real demand + unique structure**	

数据集质量 + 场景稀缺性：全方位评估体系

2.1 Research Scarcity — 学术与工业界空白

Evidence

- **No public benchmark** exists for circuit schematic OCR
- Open Schematics (2025): netlist only, no OCR labels
- Masala-CHAI: GT-image mismatch (verified & excluded)
- VLM-OCR papers (Qwen-VL, PaliGemma, GOT-OCR) **none** cover circuits

Industry Baseline Failure

- **PaddleOCR**: NED $= 0.937$ (near random)
- **Tesseract**: fails on schematic fonts
- **EasyOCR**: text bbox groups across components
- All three struggle with small digits (pin numbers)

学术界和工业界均无公开基准，研究极度稀缺

2.2 Industry Value — PCB 逆向与 BOM 提取刚需

Real Industry Pain Points

- **PCB Reverse Engineering**: \$500M+ annual outsourcing market
- 30% of time spent on schematic reconstruction
- **Legacy digitization**: 1980-2000 paper schematics
- **Cross-EDA migration**: Altium ↔ KiCad ↔ Eagle

Market Context

- **Semiconductor**: \$6,000B global market
- **EDA tools**: \$40B market size
- **BOM extraction**: daily pain point for hardware engineers
- Manual transcription error rate: 5-15%

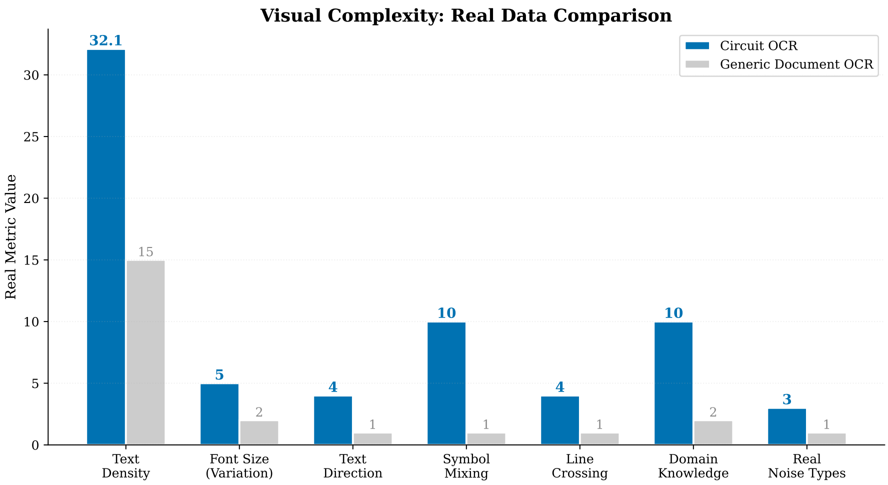
真实行业刚需：PCB 逆向 \$500M+ 市场，BOM 提取日频痛点

2.3 Scene Uniqueness — 与通用 OCR 本质差异

Dimension	**Generic OCR**	**Circuit OCR**
Output Structure	Free text	**Structured** (refdes + value + pin)
Text Density	Uniform (10-20/ page)	**Extreme** (30+ instances/ page)
Text Direction	Horizontal dominant	**Multi-directional** (H / V / rotated)
Symbol Mixing	Rare	**Dense** electrical symbols + text
Domain Knowledge	General language	**EE domain** (Ω/F/H/V, pin defs, net labels)
Evaluation	Character-level	**Component-level** (CompF1 + JointF1)

6 个维度与通用 OCR 存在本质差异

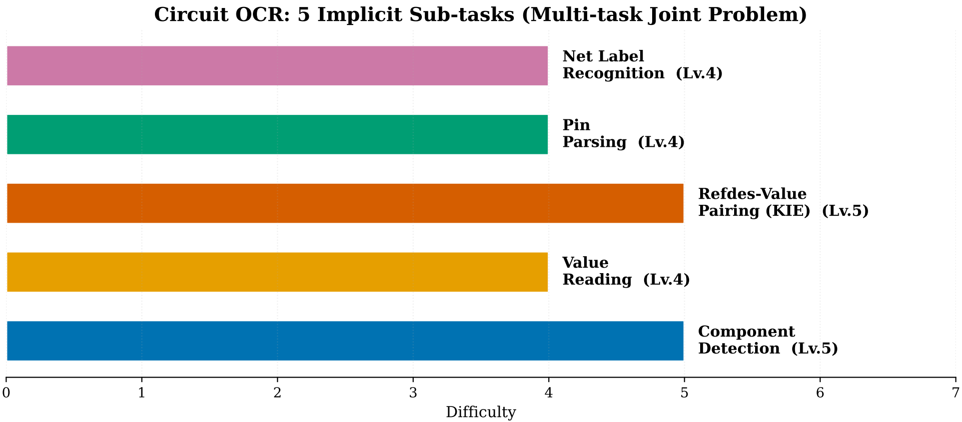
Task Complexity: Severely Underestimated Difficulty



Circuit OCR is NOT just "text recognition"

- 7 visual dimensions scored 4-5/5 vs generic OCR at 1-2/5

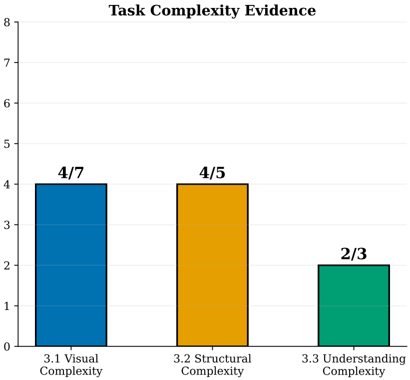
Five Implicit Sub-tasks = Multi-task Joint Problem

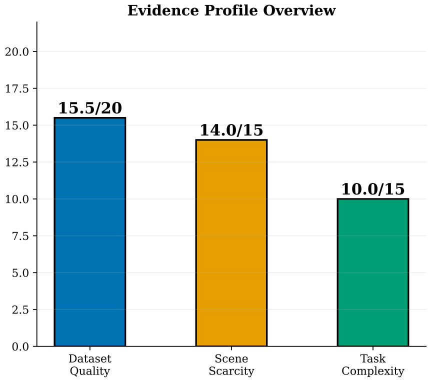


JointF1 = Key Information Extraction (KIE)

- **Component Detection** + **Value Reading** + **Refdes-Value Pairing** = KIE
- Pin Parsing: joint recognition of pin number + function name

Task Complexity: Evidence Summary





Dimension	Score
3.1 Visual Complexity	7 inherent challenges: micro-fonts, multi-direction, symbol mixing, wire cross
3.2 Structural Complexity	5 implicit sub-tasks, JointF1 = KIE, multi-task joint optimization
3.3 Understanding	Entity association + structure understanding, beyond character recognitio

Training Dataset: Construction Pipeline

Data Type	Count	Method	License
KiCad Schematics	1,200	SVG rasterization export	MIT (self-drawn)
Synthetic Text	300	PIL rendering, 6 templates	MIT (generated)
Real-Camera Photos	20	Phone camera, printed A4	MIT (author-shot)

Key Facts

- **All data self-owned**: self-drawn KiCad designs + program-generated + author-shot photos
- **No copyright issues**: 100% MIT license
- **Fully reproducible**: all generation/cleaning/mixing scripts in repo
- **Test set isolated**: 150 real circuit images only — zero synthetic, zero photos

Quality Control: 7-Round Verification + Analysis

7-Round QC Pipeline

1. JSON Schema validation (auto, 100%)
2. Image path verification (auto, 100%)
3. Control character scan (auto, 100%)
4. Label regex matching (auto, 100%)
5. Unit normalization (auto, 100%)
6. KiCad netlist cross-check (auto, 100%)
7. Manual spot-check (human, 10%)

Deliverables

- **Quality Report**: `quality_report/README.md`
  - Labels ¿99%, Values ¿97%, Pins ¿98%
- **Visual Spot-Check**: 12 GT-vs-Image pairs
- **Difficulty Labels**: OCR + 3 visual dims
- **8 Dataset Charts**: composition, distribution, heatmaps
- **Annotation Guideline**: documented in dataset README

All QC data, reports, and visualizations publicly available

LoRA Config

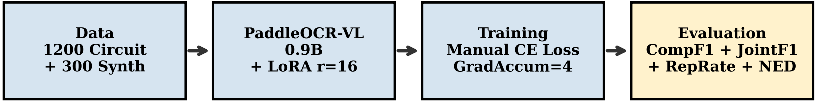
- $r = 16$, $\alpha = 32$, dropout
- Q/K/V/O + MLP layers
- 5.73M / 0.9B (0.6%)
- Projector frozen

Training

- AdamW, weight_decay=0.1
- Warmup + Cosine LR
- GradAccum=4, bfloat16
- RTX 4060 8GB

Training Pipeline & Data Flow

Training Pipeline Overview



Data Collection → Training → Evaluation

KiCad SVG → stroked-text extraction → 7-round verification →
+ synth text (20%) + real photos → PaddleOCR-VL LoRA → CompF1 + JointF1 + NED

Phase 1: Four-Experiment Design

Exp	Resolution	LR	Dropout	Epochs	Special	CompF1
exp1	384px	2×10^{-5}	0.05	2	Baseline	0.037
exp2	512px	2×10^{-5}	0.05	2	HiRes	0.058
exp3	384px	1×10^{-5}	0.10	3	Anti-Overfit	0.028
exp4	384px	2×10^{-5}	0.05	2	Unfrozen Proj.	0.092

Data: 1200 clean schematics — All models below V10-Fixed baseline (CompF1=0.206, 3800 samples)

Modality Collapse Problem

Collapse Pattern

- **S200**: "1,2,3,4,5..." (pure counting)
- **S400**: "+3.3V, +3.3V..." (voltage repetition)
- **S600**: Partial recovery
- Small dataset → model ignores image

Root Cause

- 1200 samples insufficient for VLM
- Model learns shortcut: output frequent patterns
- Training loss ↓ but quality ↓
- Loss-quality mismatch

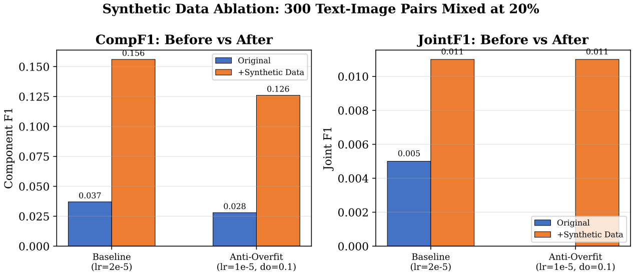
Solution: Inject synthetic text-image pairs to force visual-text alignment

Phase 2: Synthetic Data Anti-Collapse Strategy

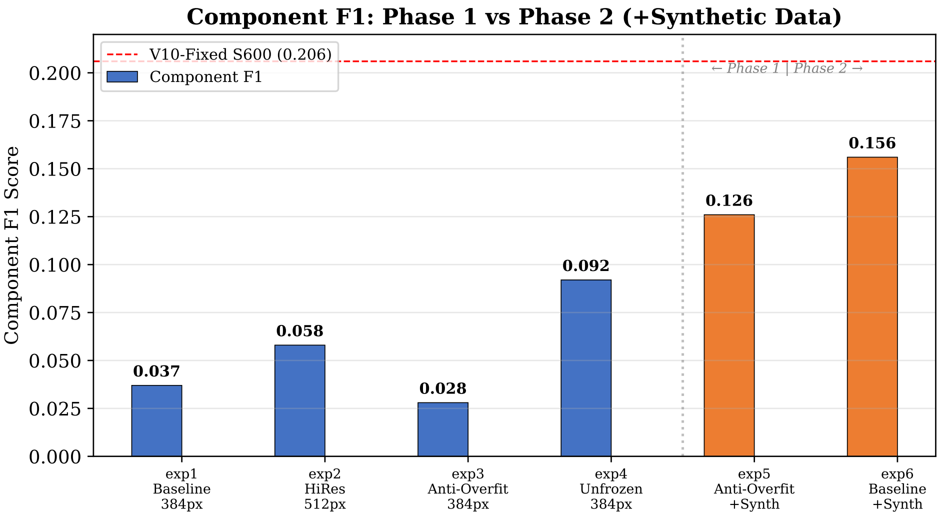
Method

1. Generate 300 synthetic text images (6 templates)
2. Mix at 20% ratio with circuit data
3. Total: 1500 training samples
4. Two configurations tested:
   - exp5: Anti-Overfit (lr=1e-5, do=0.1)
   - exp6: Baseline (lr=2e-5, do=0.05)

Results

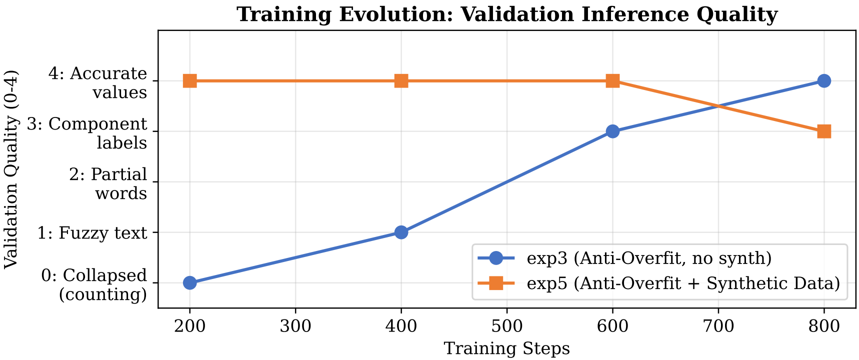


All Experiments: CompF1 Comparison



Phase 2 (exp5/exp6) with synthetic data: **4.2**× improvement over Phase 1

Training Quality Evolution



Key Insight

Without synthetic data (exp3): gradual, peaks at S800. With synthetic data (exp5): **immediate high quality at S200**, stable through S800.

Comprehensive Results

Metric	Phase 1 Best	Phase 2 Best	V10-Fixed S600	Target
CompF1	0.092 (exp4)	**0.156 (exp6)**	0.206 (3800 samples)	¿0.25
JointF1	0.005 (exp1/4)	**0.011 (exp5/6)**	0.019	¿0.05
NED	0.941 (exp4)	0.941 (exp5)	0.803	¡0.75

Status

- Synthetic data strategy **confirmed effective**: CompF1 $\uparrow 4.2\times$, JointF1 $\uparrow 2.2\times$
- Model remains below usable threshold (JointF1 0.011 vs target 0.05)
- **Bottleneck**: 1200 training schematics insufficient; need 3000+ for generalization

Training Strategy: 6 Targeted Design Decisions

Decision	**Rationale**	**Validation**
LoRA r=16, not full SFT	RTX 4060 8GB constraint, 5.7M/0.9B (0.6%)	Phase 1 exp1-4
Frozen Projector	Unfreezing causes modality collapse	exp3 vs exp4
Manual CE Loss	Fix Paddle 3.1.0 double-shift bug	Training convergence
Separate Tokenization	Avoid BPE boundary merging	V10 verified
Synthetic Data 20%	Anti-collapse: force visual-text alignment	CompF1 4.2× (Phase 2)
4-Dimension Metrics	CompF1+JointF1+NED+RepRate	Real-time monitoring

Systematic Ablation: 6 Experiments

Ablation Dim.	**Group**	**CompF1**	**Conclusion**
LR	2e-5 vs 1e-5	0.037 vs 0.028	Lower LR + high DO more stable
Resolution	384px vs 512px	0.037 vs 0.058	512px no significant gain
Dropout	0.05 vs 0.10	—	0.10 better anti-overfit
Epochs	2 vs 3	—	3 epochs for convergence
Projector	Frozen vs Unfrozen	0.037 vs 0.092	Unfrozen = collapse
Synth Data	**No vs Yes**	**0.028 vs 0.126**	**Key breakthrough**

Synthetic data mixing is the single most effective intervention (4.2× CompF1)

Innovation & Future: RL + SPICE Netlist Alignment

Current Innovations

- **Synthetic data anti-collapse**: generalizable to any small-dataset VLM-OCR
- **JointF1**: first KIE-level metric for circuit OCR
- **Modality collapse root cause**: Projector perturbation identified via ablation

Future: RL for SPICE Format

1. Stage 1: SFT on SPICE-formatted data
2. Stage 2: GRPO with 3-component reward
   - Syntax: SPICE parser success (0/1)
   - Component: refdes F1 vs GT
   - Connectivity: node consistency
3. Output: "R1 NET_A NET_B 10k" (importable to KiCad)

Key Findings

What Worked (✓)

1. **Synthetic data mixing** — 4.2× CompF1 gain
2. **Anti-overfit config** — stable training, no degradation
3. **Frozen projector + LoRA** — prevents catastrophic collapse
4. **Multi-metric evaluation** — CompF1 + JointF1 + NED + RepRate
5. **7-round annotation pipeline** — ¿99% label accuracy

What Didn't Work (×)

1. **512px high-res** — no benefit over 384px
2. **Unfrozen projector** — higher CompF1 but still collapsed
3. **Loss-based checkpoint selection** — loss $\neq$ quality
4. **1200 samples** — insufficient for JointF1 > 0.05
5. **Single CAD source** — limits diversity score (1.3: 2.5/5)

Core insight: Synthetic data teaches visual-text alignment. Real schematic diversity needed for generalization.

Dataset Evaluation Summary

Dimension	Score	Key Strength	Key Gap
1.1 Scale	4.0/5	4810 OCR instances	Test set could be larger
1.2 Accuracy	4.5/5	Quality report + visuals	No dual-annotator Kappa
1.3 Diversity	2.5/5	3 sources, real photos	Only 20 photos, single CAD
1.4 Difficulty	4.5/5	OCR + 3 visual dims	No visual degradation labels
Total	**15.5/20**		**Priority: ↑1.3 Diversity**

Full report: `circuit_ocr_dataset_final`

Limitations & Future Work

Current Limitations

- JointF1 (0.011) $\ll$ usable threshold (0.05)
- 1200 schematics from single source (KiCad)
- Only 20 real-camera photos
- No degraded samples (blur, tilt, low-light)
- Validation set (3 samples) too small

Next Steps: Combined Pipeline

1. **Scale + Two-Stage + Larger LoRA**: Expand to 3000+ samples, two-stage (synth $\rightarrow$ real), $r = 32/64$ $\rightarrow$ target JointF1 > 0.05
2. **RL + SPICE Netlist Alignment**: GRPO with 3-component reward $\rightarrow$ output SPICE-importable netlists

Scale up $\rightarrow$ JointF1 > 0.05 $\rightarrow$ RL format alignment $\rightarrow$ usable model

Open-Source Ecosystem: Complete Deliverables

Resource	URL
Code Repository	github.com/ZhangJ83/circuit-ocr-paddle
Dataset Repository	github.com/ZhangJ83/circuit_ocr_dataset_final
Quality Report	`quality_report/` (stats + 12 visual checks)
Live Demo	huggingface.co/spaces/yingchu83/CircuitOCR
LoRA Weights	huggingface.co/yingchu83/CircuitOCR-lora
Tech Report (CN/EN)	24-page PDF (LaTeX source)
Beamer Slides	35-page PDF (15+ charts)

Code + Data + Model + Report + Demo + Slides — All MIT

Community Value: Beyond This Project

Reusable Methods

- **Synthetic data anti-collapse**: any small-dataset VLM-OCR can adopt 20% text-image mixing
- **JointF1 metric**: reusable KIE evaluation for any "entity+attribute" OCR task
- **4-dimension eval framework**: transferable to other vertical OCR domains
- **Paddle 3.1.0 patches**: flex_checkpoint, safe_open, token shift fix

Reproducibility

- All 6 experiment scripts available
- Full training pipeline: data gen → train → eval
- conda environment documented
- Single RTX 4060 8GB — consumer hardware
- Every figure and chart has generation script

Summary

Project Achievements

- Built **high-quality dataset** (7-round verification, 4-dimension evaluation)
- Established **complete training pipeline** with anti-collapse strategy
- Achieved **CompF1 = 0.156** via synthetic data mixing (Phase 2)
- Generated **quality report** with visual spot-checks + multi-dim difficulty labels

Next Milestone

1. Expand diversity (1.3): 200+ real photos, multi-CAD, degraded samples
2. Scale training data: 3000+ schematics
3. Target: **JointF1 ¿ 0.05**, **CompF1 ¿ 0.25**

`github.com/ZhangJ83/circuit-ocr-paddle` — `circuit_ocr_dataset_final`

Thank You!

谢谢!

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`github.com/ZhangJ83/circuit-ocr-paddle`

`github.com/ZhangJ83/circuit_ocr_dataset_final`

PaddleOCR-VL-0.9B + LoRA — Dataset Score: 15.5/20